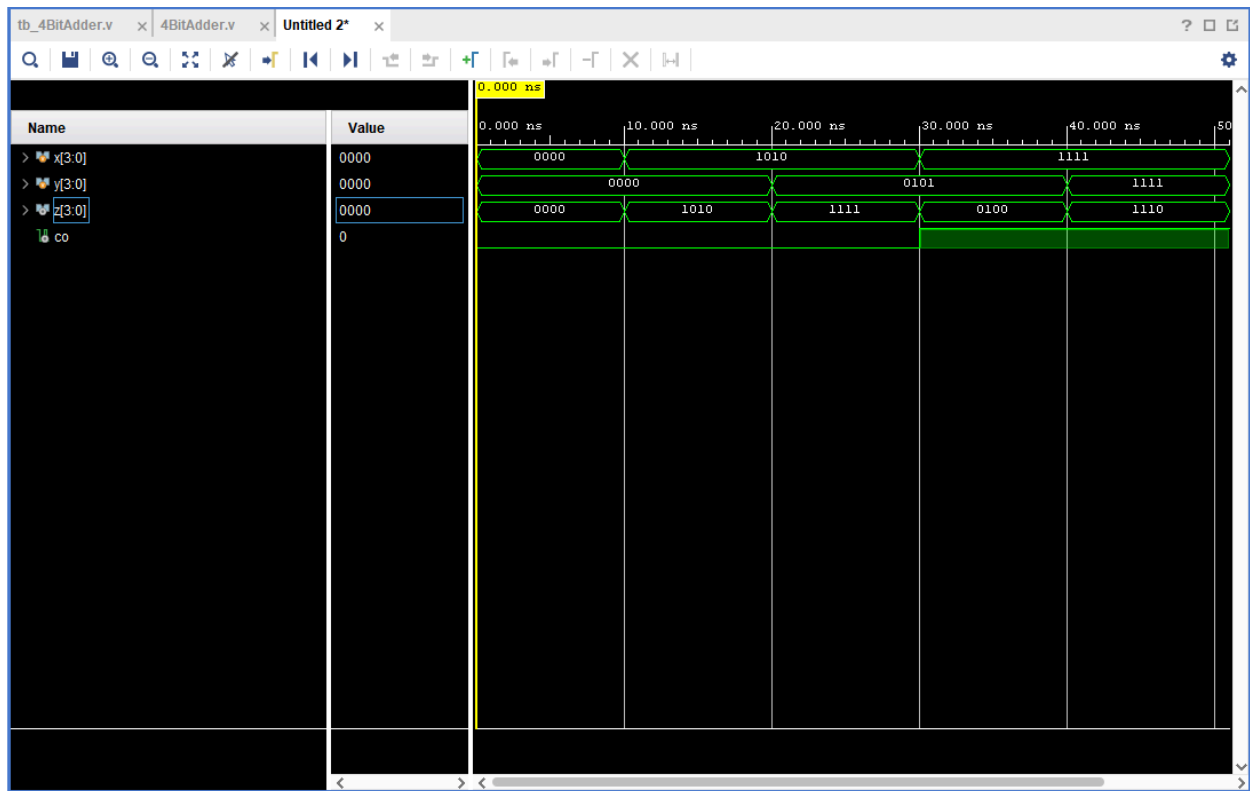


Rhilo Sotto

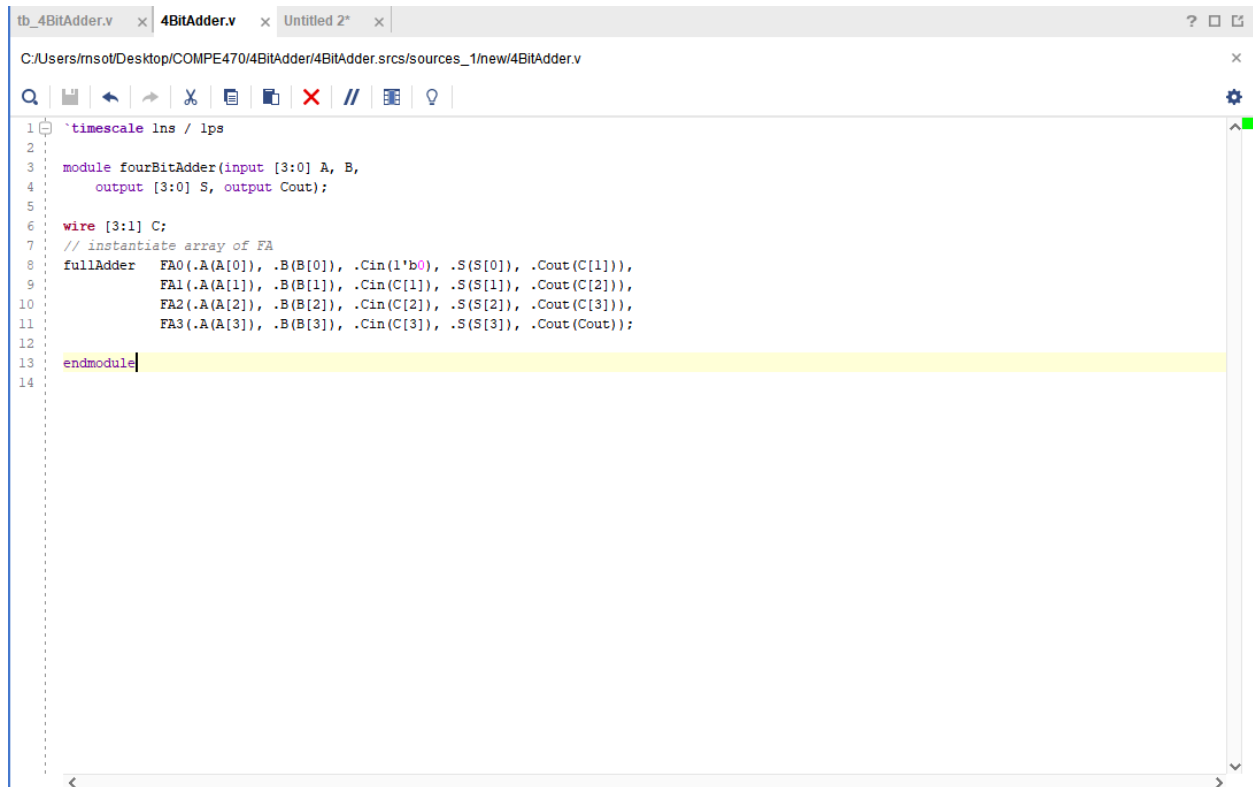
23 September 2024

Assignment 0

Waveform



Module Code



The screenshot shows a Verilog code editor with the following code:

```
1 `timescale 1ns / 1ps
2
3 module fourBitAdder(input [3:0] A, B,
4   output [3:0] S, output Cout);
5
6 wire [3:1] C;
7 // instantiate array of FA
8 fullAdder FA0(.A(A[0]), .B(B[0]), .Cin(1'b0), .S(S[0]), .Cout(C[1])),
9   FA1(.A(A[1]), .B(B[1]), .Cin(C[1]), .S(S[1]), .Cout(C[2])),
10  FA2(.A(A[2]), .B(B[2]), .Cin(C[2]), .S(S[2]), .Cout(C[3])),
11  FA3(.A(A[3]), .B(B[3]), .Cin(C[3]), .S(S[3]), .Cout(Cout));
12
13 endmodule
14
```

```
`timescale 1ns / 1ps
```

```
module fourBitAdder(input [3:0] A, B,  
    output [3:0] S, output Cout);
```

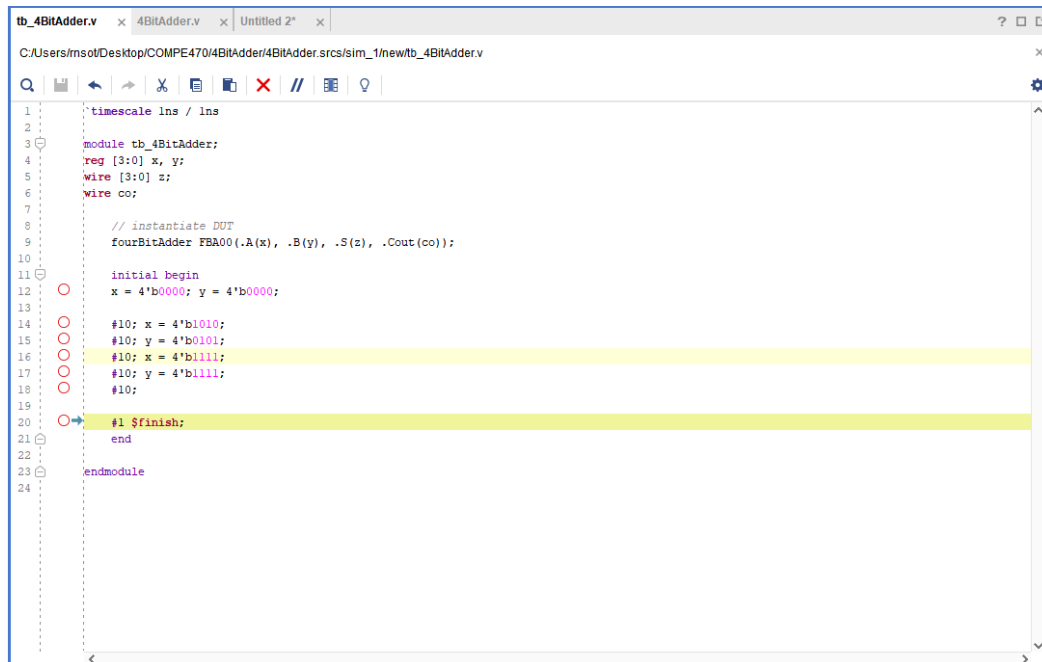
```
wire [3:1] C;
```

```
// instantiate array of FA
```

```
fullAdder FA0(.A(A[0]), .B(B[0]), .Cin(1'b0), .S(S[0]), .Cout(C[1])),  
    FA1(.A(A[1]), .B(B[1]), .Cin(C[1]), .S(S[1]), .Cout(C[2])),  
    FA2(.A(A[2]), .B(B[2]), .Cin(C[2]), .S(S[2]), .Cout(C[3])),  
    FA3(.A(A[3]), .B(B[3]), .Cin(C[3]), .S(S[3]), .Cout(Cout));
```

```
endmodule
```

Testbench Code



```
1  `timescale 1ns / 1ns
2
3  module tb_4BitAdder;
4      reg [3:0] x, y;
5      wire [3:0] z;
6      wire co;
7
8      // instantiate DUT
9      fourBitAdder FBA00(.A(x), .B(y), .S(z), .Cout(co));
10
11  initial begin
12      x = 4'b0000; y = 4'b0000;
13
14      #10; x = 4'b1010;
15      #10; y = 4'b0101;
16      #10; x = 4'b1111;
17      #10; y = 4'b1111;
18      #10;
19
20      #1 $finish;
21  end
22
23 endmodule
24
```

```
`timescale 1ns / 1ns
```

```
module tb_4BitAdder;
```

```
reg [3:0] x, y;
```

```
wire [3:0] z;
```

```
wire co;
```

```
    // instantiate DUT
```

```
    fourBitAdder FBA00(.A(x), .B(y), .S(z), .Cout(co));
```

```
    initial begin
```

```
        x = 4'b0000; y = 4'b0000;
```

```
        #10; x = 4'b1010;
```

```
        #10; y = 4'b0101;
```

```
        #10; x = 4'b1111;
```

```
        #10; y = 4'b1111;
```

```
        #10;
```

```
        #1 $finish;
```

```
    end
```

```
endmodule
```

Questions/Concerns

Not sure if this is enough testcases in my testbench, but my testbench does display all necessary aspects of a 4 bit adder; the 4 bit sum and last carry out bit being the outputs, given two 4 bit inputs